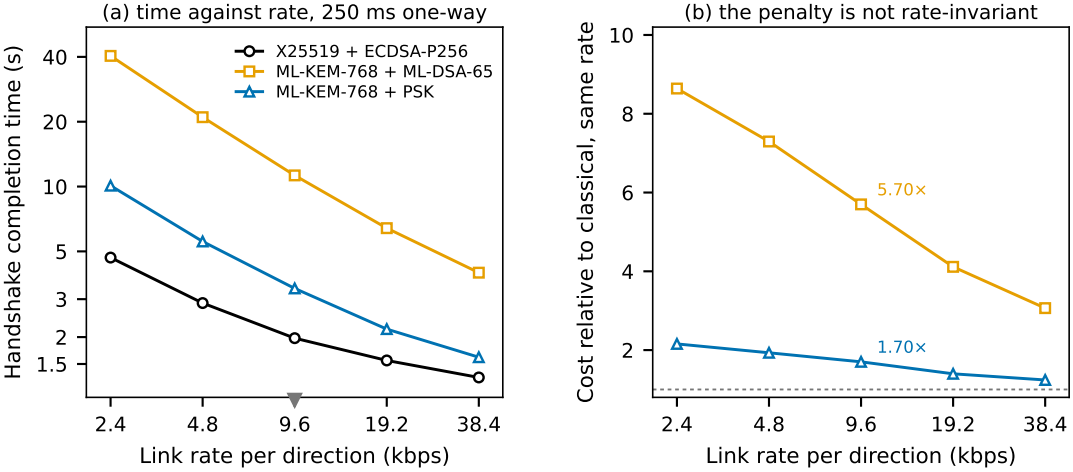
(a) time against rate, 250 ms one-way
X25519 + ECDSA-P256
ML-KEM-768 + ML-DSA-65
ML-KEM-768 + PSK
Handshake completion time (s)
40
20
10
5
3
2
1.5
2.4
4.8
9.6
19.2
38.4
Link rate per direction (kbps)
(b) the penalty is not rate-invariant
Cost relative to classical, same rate
10
8
6
4
2
5.70×
1.70×
2.4
4.8
9.6
19.2
38.4
Link rate per direction (kbps)